

Mechanical Watch Escapement Dynamics

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1 Introduction

The crux of any watch is the ability to keep time. In digital watches, maintaining constant ticks is simply a matter of timed electrical signals that can be programmed into the watch. The mechanical watch, however, is a true marvel of engineering. Within each mechanical watch is an escapement system that includes tiny, precise components that leverage the power from two springs, one of which must be meticulously calibrated. In this report, I will dive into the mechanics of the escapement system.

2 Overview of Escapement

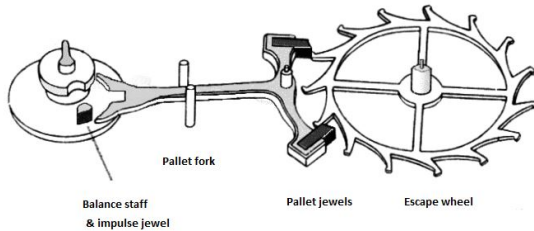


Figure 1: Overview of Escapement

In any escapement system, there exists a rotational oscillating balance staff with a small impulse jewel attached to its surface. This staff is powered below by a hairspring. During oscillation, the impulse jewel will deliver an impulse onto the pallet fork which will release the pallet jewels from their locked positions along the escape wheel. The escape wheel is connected to the mainspring which is essentially the power behind any mechanical watch. The mainspring plays the crucial role of providing constant moment to the escape wheel. When the pallet jewels are released, the escape wheel spokes will actually provide an astonishingly precise amount of energy back through the pallet fork which

will subsequently provide the impulse jewel and balance staff with just enough energy to reach the same maximum angular displacement as it started at. The duration of the interaction between impulse jewel, pallet fork, and escapement wheel is close to instantaneous (about the duration of the audible tick from the watch) and thus the oscillations of the balance staff are effectively considered simple harmonic. For further reference, watch [this video](#).

3 Period of Oscillation

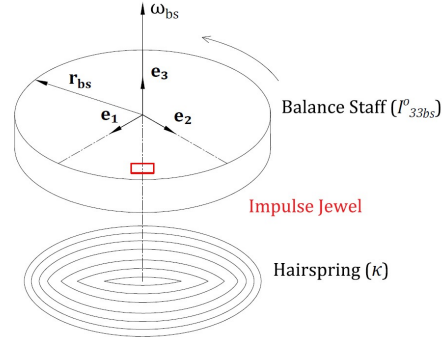


Figure 2: Balance Staff Rotation

The oscillation of the balance staff, powered by the hairspring, must keep a constant period in order to be effective. Although the pallet fork interferes with perfect simple harmonic oscillation, the duration of the interference is almost instantaneous and is thus negligible. From the second-order differential equation for simple harmonic motion $\ddot{\theta} + \omega^2 \theta = 0$ where $\omega = \sqrt{\kappa/I_{33bs}^o}$, it follows that the period of oscillation

$$T = 2\pi/\omega = 2\pi\sqrt{I_{33bs}^o/\kappa}$$

As shown, the period of the balance staff oscillations are dependent upon the moment of inertia of the balance staff as well as the κ value of the hairspring.

Of these two values, the hairspring's κ can be adjusted by a professional. This procedure is manual watch calibration.

Additionally, the maximum angular velocity of the balance staff during oscillation is given by

$$\omega_{bs_{max}} = \theta_{max} \sqrt{I_{33bs}^o / \kappa}$$

This will prove useful when analyzing Pallet Fork Dynamics.

4 Pallet Fork Dynamics

There are two significant time intervals during Pallet Fork motion. The dynamics of the fork are extremely complicated and therefore, a couple assumptions will be made for the sake of simplicity.

1. The fork is geometrically symmetrical (this is typically not the case)
2. Friction at the pivot of the fork are negligible
3. Impulsive forces delivered to the fork by the impulse jewel will be set as constant forces equal to

$$\bar{F}_{jwl} = \frac{\int F_{jwl}(t) dt}{\Delta t}$$

4. All other external forces are assumed constant

4.1 Timestamp 1 denoted: subscript 1

During the first timestamp, the impulse jewel bumps into the left prong of the fork delivering an impulse which overcomes frictional forces between the escape wheel's spoke and the fork's escapement jewel. The source of the normal force F_N that causes F_{fr} is the mainspring applying a constant moment on the escape wheel.

By definition of moment

$$\begin{aligned} M_{jwl1}^o &= r_1 \times F_{jwl1} = F_{jwl1} r_1 e_3 \\ M_{fr}^o &= r_2 \times F_{fr} = -F_{fr} r_2 e_3 \end{aligned}$$

By Euler's Law

$$\begin{aligned} \sum M^o &= \dot{H}_f^o \\ \Rightarrow M_{jwl1}^o - M_{fr}^o &= I_{33f}^o \dot{\omega}_{f1} = I_{33f}^o \alpha_{f1} \end{aligned}$$

Solve for angular acceleration of the fork

$$\alpha_{f1} = \frac{M_{jwl1}^o - M_{fr}^o}{I_{33f}^o} = \frac{\bar{F}_{jwl1} r_1 - F_{fr} r_2}{I_{33f}^o} e_3$$

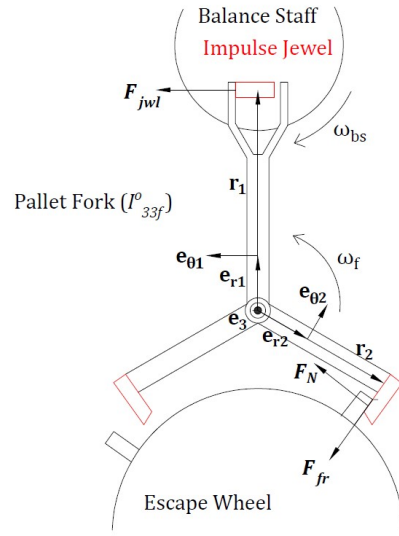


Figure 3: Free Body Diagram during initial impact

Using α_{f1} and Δt_1 (the time elapsed during initial impact), solve for angular velocity of the fork

$$\omega_{f1} = \alpha_{f1} \Delta t_1 = \frac{\bar{F}_{jwl1} r_1 - F_{fr} r_2}{I_{33f}^o} \Delta t_1 e_3$$

Since the linear velocity of the impact jewel and the fork prongs are the same, it follows that $\omega_{bs1} = \frac{r_1}{r_{bs}} \omega_{f1}$. Using this relationship derive

$$\begin{aligned} \omega_{f1} &= \frac{\bar{F}_{jwl1} r_1 - F_{fr} r_2}{I_{33f}^o} \Delta t_1 \\ \omega_{bs1} &= \frac{\bar{F}_{jwl1} r_1 - F_{fr} r_2}{I_{33f}^o} \Delta t_1 * \frac{r_1}{r_{bs}} \end{aligned}$$

where ω_{f1} denotes the angular velocity of the fork and ω_{bs1} denotes the angular velocity of the balance staff at the end of timestep 1.

4.2 Timestamp 2 denoted: subscript 2

During the second timestamp, a much more subtle exchange occurs. The impulse jewel has slowed down due to the initial impact and thus needs to speed back up to its initial impact velocity. The right fork prong thus needs to deliver an impulse to the jewel, causing it to accelerate. Let ω_{bs2} and ω_{f2} denote the angular velocities of the balance staff and fork at the end of timestamp 2. It is necessary to set $\omega_{bs2} = \omega_{bs_{max}} = \theta_{max} \sqrt{\kappa / I_{33bs}^o}$ and subsequently $\omega_{f2} = \frac{r_{bs}}{r_1} \omega_{bs2} = \frac{r_{bs}}{r_1} \theta_{max} \sqrt{\kappa / I_{33bs}^o}$

Similar to timestamp 1, apply Euler's Law using the new moments and solve for angular acceleration.

$$\sum M_f^o = \dot{H}_f^o$$

$$\Rightarrow M_{ms}^o - M_{jwl_2}^o = I_{33f}^o \dot{\omega}_f = I_{33f}^o \alpha_{f_2}$$

$$\alpha_{f_2} = \frac{M_{ms}^o - M_{jwl_2}^o}{I_{33f}^o} = \frac{F_{ms}r_2 - \bar{F}_{jwl_2}r_1}{I_{33f}^o} e_3$$

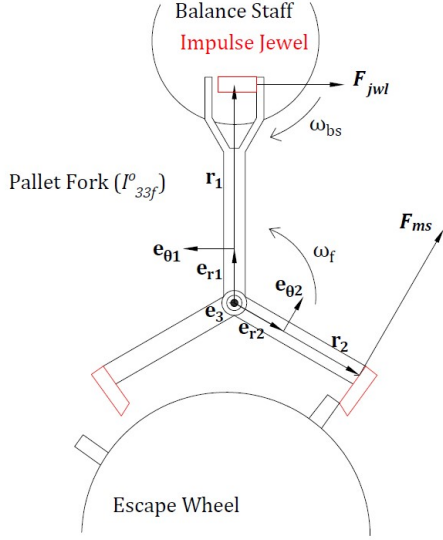


Figure 4: Free Body Diagram during secondary impact

Using α_{f_2} and Δt_2 (the time elapsed during secondary impact), set up the equation for the fork's new angular velocity and substitute.

$$\omega_{f_2} = \omega_{f_1} + \alpha_{f_2} \Delta t_2$$

$$\Rightarrow \frac{r_{bs}}{r_1} \theta_{max} \sqrt{\kappa / I_{33bs}^o} = \frac{\bar{F}_{jwl_1} r_1 - F_{fr} r_2}{I_{33f}^o} \Delta t_1 + \frac{F_{ms} r_2 - \bar{F}_{jwl_2} r_1}{I_{33f}^o} \Delta t_2$$

Since the change in angular velocity of the balance staff is the same across both timestamps (and the moment of inertia stays constant), the impulse delivered to the balance staff must also be the same across both timestamps. Thus it follows that $\bar{F}_{jwl_2} = \frac{\Delta t_1}{\Delta t_2} \bar{F}_{jwl_1}$.

Using this relationship along with the substituted angular velocity equation, we derive.

$$F_{ms} \Delta t_2 - F_{fr} \Delta t_1 = \frac{r_{bs}}{r_1 r_2} \theta_{max} I_{33f}^o \sqrt{\kappa / I_{33bs}^o}$$

This critical equation defines the relationship between the force provided by the mainspring F_{ms} and the force of friction between the escape wheel spokes and the pallet jewel F_{fr} .

5 Mainspring Energy Losses

Lastly, it is important to mention the energy lost by the mainspring to every set of impulses. By conservation

of energy, it is clear that the energy provided by the mainspring in timestamp 2 goes solely to increasing the angular velocity of both the balance staff and fork. Thus it follows that

$$\Delta E_{ms} = -\left(\frac{1}{2} I_{33bs}^o (\omega_{bs_2}^2 - \omega_{bs_1}^2) + \frac{1}{2} I_{33f}^o (\omega_{f_2}^2 - \omega_{f_1}^2)\right)$$

Most of the energy is dissipated on two sources.

1. Frictional losses between the pallet jewel and escape wheel spokes
2. Perfectly inelastic collision of pallet fork and stopping pegs

The energy lost by the stopper is clear. All of the rotational kinetic energy of the fork after timestamp 2 is dissipated to unusable energy after the fork collides with the stopping peg and comes to rest. Thus

$$\Delta E_{fstopper} = -\frac{1}{2} I_{33f}^o \omega_{f_2}^2$$

Since energy is dissipated through two sources, we can set up the equation $\Delta E_{ms} = \Delta E_{ffr} + \Delta E_{fstopper}$. From this equation and the expression for $\Delta E_{fstopper}$, it follows that frictional energy loss is given by

$$\Delta E_{ffr} = -\left(\frac{1}{2} I_{33bs}^o (\omega_{bs_2}^2 - \omega_{bs_1}^2) - \frac{1}{2} I_{33f}^o \omega_{f_1}^2\right)$$

6 Conclusion and Application

In conclusion, through analysis of a simplified model of escapement, two equations stand out as important for practical use. The first is the equation for balance staff oscillation frequency given by

$$\omega = \sqrt{\kappa / I_{33bs}^o}$$

which is useful when deciding how to calibrate the hairspring by adjusting its κ value. Typically, the length of the hairspring is adjusted in slight amounts which directly influence the rate at which the watch will run.

The second practical equation is the equation for mainspring energy loss given by

$$\Delta E_{ms} = -\left(\frac{1}{2} I_{33bs}^o (\omega_{bs_2}^2 - \omega_{bs_1}^2) + \frac{1}{2} I_{33f}^o (\omega_{f_2}^2 - \omega_{f_1}^2)\right)$$

This equation allows the user to determine how much energy is being released by their mainspring upon every impact of the impulse jewel. Given that mainsprings can typically hold 0.3 Joules of energy and knowing the rate at which impacts occur, watch users can use the rate of energy loss to calculate how much time elapses before they have to rewind their mainsprings.

References

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